

Foundation (Jalapeno)

Q1. Rationalize the denominator of $\frac{1}{\sqrt{2}}$.

- (A) $\frac{\sqrt{2}}{2}$
- (B) $\frac{1}{2\sqrt{2}}$
- (C) $\frac{2}{\sqrt{2}}$
- (D) $\sqrt{2}$
- (E) $\frac{1}{2}$

Q2. Which expression has a rationalized denominator?

- (A) $\frac{1}{\sqrt{3}}$
- (B) $\frac{1}{2\sqrt{3}}$
- (C) $\frac{\sqrt{3}}{3}$
- (D) $\frac{\sqrt{3}}{2\sqrt{3}}$
- (E) $\frac{1}{\sqrt{3}} \cdot \frac{\sqrt{3}}{\sqrt{3}}$

Q3. A stock's price is $25 - \sqrt{11}$. What is the price with a rationalized denominator?

- (A) $25 - \sqrt{11}$
- (B) $25 - \frac{\sqrt{11}}{1}$
- (C) $25 - \frac{\sqrt{11}\sqrt{11}}{\sqrt{11}}$
- (D) $25 - \frac{11}{\sqrt{11}}$
- (E) $25 - \frac{\sqrt{11}}{1} \cdot \frac{\sqrt{11}}{\sqrt{11}}$

Q4. If $x = \frac{2}{\sqrt{5}}$, what is x with a rationalized denominator?

- (A) $\frac{2}{\sqrt{5}}$
- (B) $\frac{2}{5}$
- (C) $\frac{\sqrt{5}}{5}$
- (D) $\frac{2\sqrt{5}}{5}$
- (E) $2\sqrt{5}$

Q5. A tax is calculated as $\frac{1}{\sqrt{7}}$ of the total income. What is the

tax with a rationalized denominator?

- (A) $\frac{1}{\sqrt{7}}$
- (B) $\frac{\sqrt{7}}{7}$
- (C) $\frac{1}{7\sqrt{7}}$
- (D) $\frac{7}{\sqrt{7}}$
- (E) $\frac{\sqrt{7}}{7}$

Q6. Rationalize the denominator of $\frac{1+\sqrt{6}}{\sqrt{6}}$.

- (A) $\frac{1}{\sqrt{6}} + \frac{\sqrt{6}}{\sqrt{6}}$
- (B) $\frac{1}{\sqrt{6}} + \frac{\sqrt{6}}{6}$
- (C) $\frac{\sqrt{6}+6}{6}$
- (D) $\frac{6+\sqrt{6}}{6}$
- (E) $\frac{1+\sqrt{6}}{6}$

Q7. A savings account earns $\frac{1}{\sqrt{3}}$ interest. What is the interest with a rationalized denominator?

- (A) $\frac{1}{\sqrt{3}}$
- (B) $\frac{\sqrt{3}}{3}$
- (C) $\frac{1}{3\sqrt{3}}$
- (D) $\frac{3}{\sqrt{3}}$
- (E) $\frac{\sqrt{3}}{3}$

Q8. The value of a stock option is $\frac{10}{\sqrt{8}}$. What is the value with a rationalized denominator?

- (A) $\frac{10}{2\sqrt{2}}$
- (B) $\frac{5}{\sqrt{2}}$
- (C) $\frac{5\sqrt{2}}{4}$
- (D) $\frac{5\sqrt{2}}{2}$
- (E) $\frac{10\sqrt{2}}{8}$

Open Ended Questions (FRQ)

Q9. Rationalize the denominator of $\frac{2+\sqrt{5}}{\sqrt{5}-1}$ and simplify.

Q10. The cost of a company's stock can be calculated using the formula $C = \frac{250}{\sqrt{t+2}}$, where t is the time in months. Rationalize the denominator and simplify.

Chef's Tips

- Read the instructions carefully.
- Use a calculator only when necessary.
- Show all work and explain your reasoning.